Module 3

NETWORKING TECHNOLOGIES

3.1. Wired: DSL, Cable Networks, SONET

3.1.1. Digital subscriber line (DSL)

Digital subscriber line (**DSL**; originally **digital subscriber loop**) is a family of technologies that are used to transmit digital data over telephone lines. In telecommunications marketing, the term DSL is widely understood to mean asymmetric digital subscriber line (ADSL), the most commonly installed DSL technology, for Internet access.

DSL service can be delivered simultaneously with wired telephone service on the same telephone line since DSL uses higher frequency bands for data. On the customer premises, a DSL filter on each non-DSL outlet blocks any high-frequency interference to enable simultaneous use of the voice and DSL services.

The bit rate of consumer DSL services typically ranges from 256 kbit/s to over 100 Mbit/s in the direction to the customer (downstream), depending on DSL technology, line conditions, and service-level implementation. Bit rates of 1 Gbit/s have been reached

Operation

Telephones are connected to the telephone exchange via a local loop, which is a physical pair of wires. The local loop was originally intended mostly for the transmission of speech, encompassing an audio frequency range of 300 to 3400 hertz (commercial bandwidth). However, as long-distance trunks were gradually converted from analog to digital operation, the idea of being able to pass data through the local loop (by using frequencies above the voice band) took hold, ultimately leading to DSL.

The local loop connecting the telephone exchange to most subscribers has the capability of carrying frequencies well beyond the 3400 Hz upper limit of POTS. Depending on the length and quality of the loop, the upper limit can be tens of megahertz. DSL takes advantage of this unused bandwidth of the local loop by creating 4312.5 Hz wide channels starting between 10 and 100 kHz, depending on how the system is configured. Allocation of channels continues to higher frequencies (up to 1.1 MHz for ADSL) until new channels are deemed unusable. Each channel is evaluated for usability in much the same way an analog modem would on a POTS

connection. More usable channels equate to more available bandwidth, which is why distance and line quality are a factor (the higher frequencies used by DSL travel only short distances).

The pool of usable channels is then split into two different frequency bands for upstream and downstream traffic, based on a preconfigured ratio. This segregation reduces interference. Once the channel groups have been established, the individual channels are bonded into a pair of virtual circuits, one in each direction. Like analog modems, DSL transceivers constantly monitor the quality of each channel and will add or remove them from service depending on whether they are usable. Once upstream and downstream circuits are established, a subscriber can connect to a service such as an Internet service provider or other network services, like a corporate MPLS network.

The underlying technology of transport across DSL facilities uses modulation of high-frequency carrier waves, an analog signal transmission. A DSL circuit terminates at each end in a modem which modulates patterns of bits into certain high-frequency impulses for transmission to the opposing modem. Signals received from the far-end modem are demodulated to yield a corresponding bit pattern that the modem passes on, in digital form, to its interfaced equipment, such as a computer, router, switch, etc.

Unlike traditional dial-up modems, which modulate bits into signals in the 300–3400 Hz audio baseband, DSL modems modulate frequencies from 4000 Hz to as high as 4 MHz. This frequency band separation enables DSL service and plain old telephone service (POTS) to coexist on the same cables. On the subscriber's end of the circuit, inline DSL filters are installed on each telephone to pass voice frequencies but filter the high-frequency signals that would otherwise be heard as hiss. Also, nonlinear elements in the phone could otherwise generate audible intermodulation and may impair the operation of the data modem in the absence of these low-pass filters. DSL and RADSL modulations do not use the voice-frequency band so high-pass filters are incorporated in the circuitry of DSL modems filter out voice frequencies.

Because DSL operates above the 3.4 kHz voice limit, it cannot pass through a loading coil, which is an inductive coil that is designed to counteract loss caused by shunt capacitance (capacitance between the two wires of the twisted pair). Loading coils are commonly set at regular intervals in POTS lines. Voice service cannot be maintained past a certain distance without such coils. Therefore, some areas that are within range for DSL service are disqualified from eligibility because of loading coil placement. Because of this, phone companies endeavor

to remove loading coils on copper loops that can operate without them. Longer lines that require them can be replaced with fiber to the neighborhood or node (FTTN).

Most residential and small-office DSL implementations reserve low frequencies for POTS, so that (with suitable filters and/or splitters) the existing voice service continues to operate independently of the DSL service. Thus POTS-based communications, including fax machines and dial-up modems, can share the wires with DSL. Only one DSL modem can use the subscriber line at a time. The standard way to let multiple computers share a DSL connection uses a router that establishes a connection between the DSL modem and a local Ethernet, powerline, or Wi-Fi network on the customer's premises.

The theoretical foundations of DSL, like much of communication technology, can be traced back to Claude Shannon's seminal 1948 paper, "A Mathematical Theory of Communication". Generally, higher bit rate transmissions require a wider frequency band, though the ratio of bit rate to symbol rate and thus to bandwidth are not linear due to significant innovations in digital signal processing and digital modulation methods.

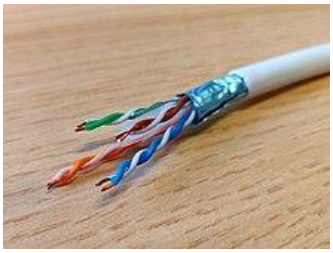
3.1.2. Cable Networks

Networking cable is a piece of networking hardware used to connect one network device to other network devices or to connect two or more computers to share devices such as printers or scanners. Different types of network cables, such as coaxial cable, optical fiber cable, and twisted pair cables, are used depending on the network's topology, protocol, and size. The devices can be separated by a few meters (e.g. via Ethernet) or nearly unlimited distances (e.g. via the interconnections of the Internet).

While wireless networks are more easily deployed when total throughput is not an issue, most permanent larger computer networks utilize cables to transfer signals from one point to another.

There are several technologies used for network connections. Patch cables are used for short distances in offices and wiring closets. Electrical connections using twisted pair or coaxial cable are used within a building. Optical fiber cable is used for long distances or for applications requiring high bandwidth or electrical isolation. Many installations use structured cabling practices to improve reliability and maintainability. In some home and industrial applications power lines are used as network cabling.

Twisted pair



Twisted pair cabling is a form of wiring in which pairs of wires (the forward and return conductors of a single circuit) are twisted together for the purposes of canceling out electromagnetic interference (EMI) from other wire pairs and from external sources. This type of cable is used for home and corporate Ethernet networks. Twisted pair cabling is used in short patch cables and in the longer runs in structured cabling.

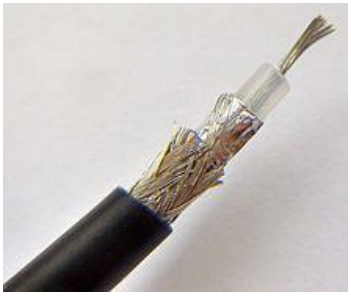
There are two types of twisted pair cables: shielded and unshielded.

Fiber optic cable



An optical fiber cable consists of a center glass core surrounded by several layers of protective material. The outer insulating jacket is made of Teflon or PVC to prevent interference. It is expensive but has higher bandwidth and can transmit data over longer distances. There are two major types of optical fiber cables: shorter-range multi-mode fiber and long-range single-mode fiber.

Coaxial cable



Coaxial cables form a transmission line and confine the electromagnetic wave to an area inside the cable between the center conductor and the shield. The transmission of energy in the line occurs totally through the dielectric inside the cable between the conductors. Coaxial lines can therefore be bent and twisted (subject to limits) without negative effects, and they can be strapped to conductive supports without inducing unwanted currents in them.

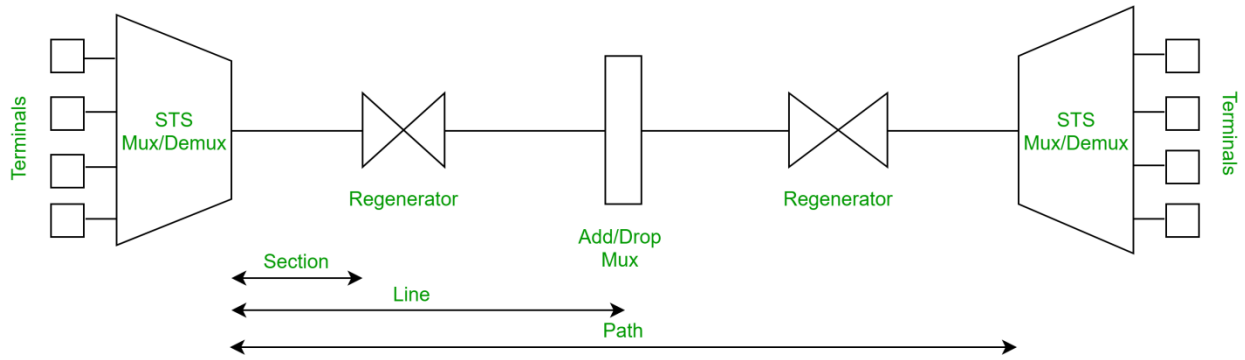
3.1.3. Synchronous Optical Network (SONET)

SONET is a standardized digital communication protocol that uses a fiber optic medium to transfer huge amounts of data across great distances. Multiple digital data streams are transmitted via optical fiber at the same time utilizing LEDs and laser beams in SONET.

Synchronous optical networking (SONET) and synchronous digital hierarchy (SDH) are established protocols that use lasers or highly coherent light from light-emitting diodes to transport multiple digital bit streams simultaneously across optical fiber (LEDs). Data can also be transmitted over an electrical link at modest transmission rates. The technique was created to replace the plesiochronous digital hierarchy (PDH) system for carrying huge volumes of voice and data traffic over a single cable without synchronization issues.

SONET and SDH were designed to transmit circuit mode communications (e.g., DS1, DS3) from a variety of different sources, but they were principally meant to enable real-time, uncompressed, circuit-switched audio encoded in PCM format. Prior to SONET/SDH, the main challenge was that the synchronization sources of these multiple circuits were all different. This meant that each circuit ran at a little different pace and had a slightly different phase. Within a single framing protocol, SONET/SDH allows for the simultaneous transmission of several distinct circuits of various origins.

Elements of SONET

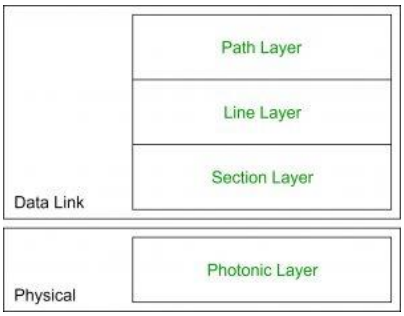


- **STS Multiplexer** − It performs signal multiplexing and transforms an electrical signal into an optical signal.
- **STS Demultiplexer** − It performs signal demultiplexing. This device converts an optical signal into an electrical signal.
- **Regenerator** − A regenerator is a repeater that takes an optical signal and strengthens it.
- **Add/Drop Multiplexer** − This device allows you to combine signals from several sources into a single route or delete a signal.

Why is SONET Used?

SONET is a technology that converts electrical signals into optical signals, allowing them to travel across larger distances.

Layers of SONET



- **Path Layer** − The Path Layer is in charge of moving signals from their optical source to their optical destination. Path layer functionalities are provided by STS Mux/Demux.
- **Line Layer** − The Line Layer is in charge of signal movement across a physical line. Line layer features are available in STS Mux/Demux and Add/Drop Mux.
- **Section Layer** − The Section Layer is in charge of signal flow throughout a physical section. Section layer functions are provided by each network device.
- **Photonic layer** − The photonic layer relates to the OSI model's physical layer. The optical fiber channel's physical characteristics (presence of light = 1 and absence of light = 0) are included.

Advantages of Using SONET

- Data is sent over long distances.
- Electromagnetic interference is minimal.
- Data rates are quite high.
- Provides broad bandwidth.

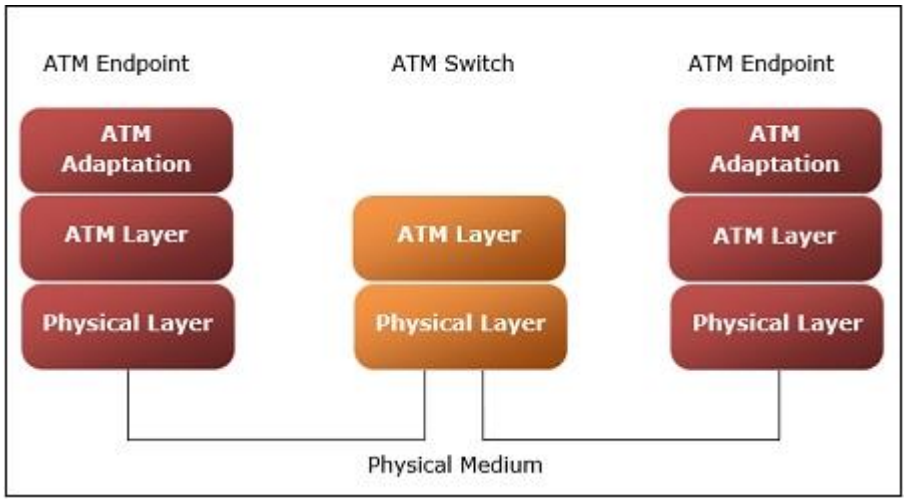
3.2. ATM, VLAN

3.2.1.ATM and ATM Networks

ATM stands for Asynchronous Transfer Mode. It is a switching technique that uses time division multiplexing (TDM) for data communications.

ATM networks are connection oriented networks for cell relay that supports voice, video and data communications. It encodes data into small fixed - size cells so that they are suitable for TDM and transmits them over a physical medium.

The size of an ATM cell is 53 bytes: 5 byte header and 48 byte payload. There are two different cell formats - user-network interface (UNI) and network-network interface (NNI). The below image represents the Functional Reference Model of the Asynchronous Transfer Mode.



Benefits of ATM Networks are

- It provides the dynamic bandwidth that is particularly suited for bursty traffic.
- Since all data are encoded into identical cells, data transmission is simple, uniform and predictable.
- Uniform packet size ensures that mixed traffic is handled efficiently.
- Small sized header reduces packet overload, thus ensuring effective bandwidth usage.
- ATM networks are scalable both in size and speed.

ATM reference model comprises of three layers

- **Physical Layer** − This layer corresponds to physical layer of OSI model. At this layer, the cells are converted into bit streams and transmitted over the physical medium. This layer has two sub layers: PMD sub layer (Physical Medium Dependent) and TC (Transmission Convergence) sub layer.

- **ATM Layer** −This layer is comparable to data link layer of OSI model. It accepts the 48 byte segments from the upper layer, adds a 5 byte header to each segment and converts into 53 byte cells. This layer is responsible for routing of each cell, traffic management, multiplexing and switching.
- **ATM Adaptation Layer (AAL)** −This layer corresponds to network layer of OSI model. It provides facilities to the existing packet switched networks to connect to ATM network and use its services. It accepts the data and converts them into fixed sized segments. The transmissions can be of fixed or variable data rate. This layer has two sub layers − Convergence sub layer and Segmentation and Reassembly sub layer.
- **ATM endpoints** − It contains ATM network interface adaptor. Examples of endpoints are workstations, routers, CODECs, LAN switches, etc.
- **ATM switch** −It transmits cells through the ATM networks. It accepts the incoming cells from ATM endpoints (UNI) or another switch (NNI), updates cell header and retransmits cell towards destination.

3.2.2.VLAN

A virtual LAN (VLAN) is a logical overlay network that groups together a subset of devices that share a physical LAN, isolating the traffic for each group.

A LAN is a group of computers or other devices in the same place -- e.g., the same building or campus -- that share the same physical network. A LAN is usually associated with an Ethernet (Layer 2) broadcast domain, which is the set of network devices an Ethernet broadcast packet can reach.

Computers on the LAN connect to the same network switch, either directly or through wireless access points (APs) connected to the same switch. Computers can also connect to one of a set of interconnected switches, such as a set of access switches that all connect up to a backbone switch. Once traffic crosses a router and engages Layer 3 (IP-related) functions, it is not considered to be on the same LAN, even if everything stays in the same building or floor. As a result, a location could have many interconnected LANs.

A VLAN, like the LAN it sits atop, operates at Layer 2 of the network, the Ethernet level. VLANs partition a single switched network into a set of overlaid virtual networks that can meet different functional and security requirements. This partitioning avoids the need to have multiple, distinct physical networks for different use cases.

The purpose of a VLAN

Network engineers use VLANs for multiple reasons, including the following:

- to improve performance
- to tighten security
- to ease administration

Improve performance

VLANs can improve performance for devices on them by reducing the amount of traffic a given endpoint sees and processes. VLANs break up broadcast domains, reducing the number of other hosts from which any given device sees broadcasts. For example, if all desktop voice over IP phones are on one VLAN and all workstations are on another, phones won't see any workstation-generated broadcast traffic and vice versa. Each can devote its network resources to relevant traffic only.

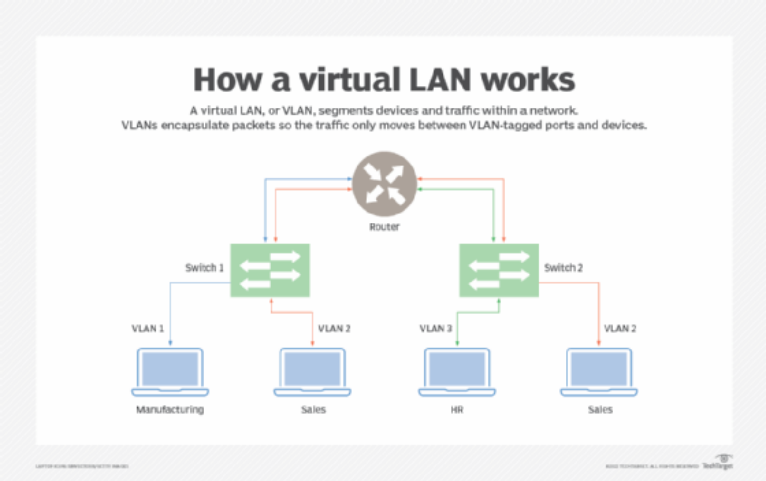
Engineers can also define different traffic-handling rules per VLAN. For example, they can set rules to prioritize video traffic on a VLAN that connects conference room equipment to help guarantee the performance of telepresence devices.

Tighten security

VLAN partitioning can also improve security by enabling a higher degree of control over which devices have access to each other. For example, network teams may restrict management access to network gear or IoT devices to specific VLANs.

Ease administration

Using VLANs to group endpoints also enables administrators to group devices for purely administrative, nontechnical purposes. For example, they may put all accounting computers on one VLAN, all human resources computers on another and so on.



Types of VLANs

VLANs can be port-based (sometimes called *static*) or use-based (sometimes called *dynamic*).

Port-based or static VLAN

Network engineers create port-based VLANs by assigning ports on a network switch to a VLAN. Those ports only communicate on the assigned VLANs, and each port is on one VLAN only. While port-based VLANs are sometimes called static VLANs, it's important to remember they aren't truly static because the VLANs assigned to the port can be changed on the fly, manually or by network automation.

Use-based or dynamic VLAN

Network engineers create use-based VLANs by assigning traffic to a VLAN dynamically, based on the traffic type or the device creating the traffic. A port might be assigned to a VLAN based on the identity of the device attached -- as indicated by a security certificate -- or on the network protocols in use. One port can be associated with multiple dynamic VLANs. Changing which device is connected through a port, or even how the existing device is used, might change the VLAN assigned to the port.

VLAN use cases

Some VLANs have simple and practical goals, such as segregating printer access. Administrators can set them up to enable computers on any given VLAN to see the printers also on that VLAN but not those outside it.

Other VLANs serve more complex purposes. For example, computers in a retail banking department cannot interact directly with computers in the trading departments. Setting up

separate VLANs for the departments is one way network engineers can enforce such segregation.

How VLANs work

A VLAN is identified on network switches by a VLAN ID. Each port on a switch can have one or more VLAN IDs assigned to it and will land in a default VLAN if no other one is assigned. Each VLAN provides data-link access to all hosts connected to switch ports configured with its VLAN ID.

A VLAN ID is translated to a VLAN tag, a 12-bit field in the header data of every Ethernet frame sent to that VLAN. Because a tag is 12 bits long, up to 4,096 VLANs can be defined per switching domain. VLAN tagging is defined by IEEE in the 802.1Q standard.

When an Ethernet frame is received from an attached host, it has no VLAN tag. The switch adds the VLAN tag. In a static VLAN, the switch inserts the tag associated with the ingress port's VLAN ID. In a dynamic VLAN, it inserts the tag associated with that device's ID or the type of traffic it generates.

Switches forward tagged frames toward their destination media access control address, forwarding only to ports with which the VLAN is associated. Broadcast, unknown unicast and multicast traffic is forwarded to all ports in the VLAN. Trunk links between switches know which VLANs span the switches, accepting and passing along all traffic for any VLAN in use on both sides of the trunk. When a frame reaches its destination switch port, the VLAN tag is removed before the frame is transmitted to the destination device.

Spanning Tree Protocol (STP) is used to create loop-free topology among the switches in each Layer 2 domain. A per-VLAN STP instance can be used, which enables different Layer 2 topologies. A multi-instance STP can also be used to reduce STP overhead if the topology is the same among multiple VLANs.

Disadvantages of VLANs

VLANs help control broadcast traffic, tighten security, ease administration and improve performance. But they have some disadvantages, too.

3.4. Cellular Networks: Introduction-Wireless links and Network characteristics –CDMA

Computer networks that are not connected by cables are called wireless networks. They generally use radio waves for communication between the network nodes. They allow devices to be connected to the network while roaming around within the network coverage.



Types of Wireless Networks

- Wireless LANs − Connects two or more network devices using wireless distribution techniques.
- Wireless MANs − Connects two or more wireless LANs spreading over a metropolitan area.
- Wireless WANs − Connects large areas comprising LANs, MANs and personal networks.

Advantages of Wireless Networks

- It provides clutter-free desks due to the absence of wires and cables.
- It increases the mobility of network devices connected to the system since the devices need not be connected to each other.
- Accessing network devices from any location within the network coverage or Wi-Fi hotspot becomes convenient since laying out cables is not needed.
- Installation and setup of wireless networks are easier.

- New devices can be easily connected to the existing setup since they needn't be wired to the present equipment. Also, the number of equipment that can be added or removed to the system can vary considerably since they are not limited by the cable capacity. This makes wireless networks very scalable.
- Wireless networks require very limited or no wires. Thus, it reduces the equipment and setup costs.

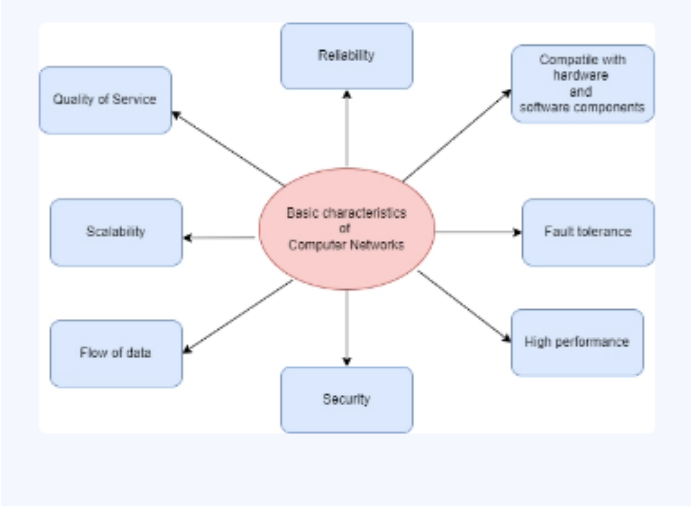
Examples of wireless networks

- Mobile phone networks
- Wireless sensor networks
- Satellite communication networks
- Terrestrial microwave networks

3.4.1.Network characteristics

A **computer network** is a grouping of two or more devices linked by a medium to exchange data. We can easily send and receive data to and from a computing device using a computer network. In this article, we are going to discuss some basic characteristics of Computer Networks.

Basic characteristics of a Computer Network



Security

Security is one of the most essential characteristics of a computer network. Most businesses nowadays rely on computers, which are accessed through networking. As a result, if computer network technology is not robust and secure, unauthorized access to the company's critical data might be possible. However, nowadays, computer networking tools primarily provide the highest level of security and prevent any unauthorized access.

Reliability

Computer networks are very reliable tools, and users can easily interconnect their devices with their help. Computer networks have alternative sources of supply to provide **high reliability**. Their networking experience remains consistent even if users need to print, check messages, attend a meeting, or access data from another computer.

Scalability

Scalability refers to the ability to scale up in response to changing needs while maintaining high performance. The internet is the best example of scalability; many new users connect via the internet and communicate with other devices, but our network still works properly.

Flow of Data

Users can access and transmit data such as files, documents, and other types of information with the help of computer networking. It is a crucial feature of a computer network as it allows data to flow from one device to another.

High performance

The performance is evaluated using the time required by a command. If it requires less time to transfer the data and the response is quick, it is a significant advantage for the users to transfer data and use multiple resources. The performance can be increased using multiple processors.

Fault tolerance

Fault tolerance is also a great feature offered by computer networks. Let’s suppose two devices are connected with both wired and wireless mediums. If the sender’s device sends the information, and the receiver’s device has a blockage at its wireless medium. It will find the other best alternative medium to send the information to the receiver’s device, which is a wired

medium in this case. Using this, they can continue to work even if the networking is down or damaged. This is how fault tolerance works.

Quality of Service (QoS)

It means users can prioritize and customize their data transmission while also minimizing any transmission delays. It also allows for the loss of any data if it occurs regularly. As a result, another feature of a computer network is that it provides a high level of service to its users.

Compatible With Hardware and Software Components

Another great feature of computer networking is allowing multiple connected devices to access the same software. It entails that the same software can be used on various devices. It improves compatibility, allowing this work to be completed. As a result, computer networks make it easier to use software and make the best use of physical components.

3.4.2. Code Division Multiple Access (CDMA)

Code Division Multiple Access (CDMA) is a sort of multiplexing that facilitates various signals to occupy a single transmission channel. It optimizes the use of available bandwidth. The technology is commonly used in ultra-high-frequency (UHF) cellular telephone systems, bands ranging between the 800-MHz and 1.9-GHz.

CDMA Overview

Code Division Multiple Access system is very different from time and frequency multiplexing. In this system, a user has access to the whole bandwidth for the entire duration. The basic principle is that different CDMA codes are used to distinguish among the different users.

Techniques generally used are direct sequence spread spectrum modulation (DS-CDMA), frequency hopping or mixed CDMA detection (JDCDMA). Here, a signal is generated which extends over a wide bandwidth. A code called **spreading code** is used to perform this action. Using a group of codes, which are orthogonal to each other, it is possible to select a signal with a given code in the presence of many other signals with different orthogonal codes.

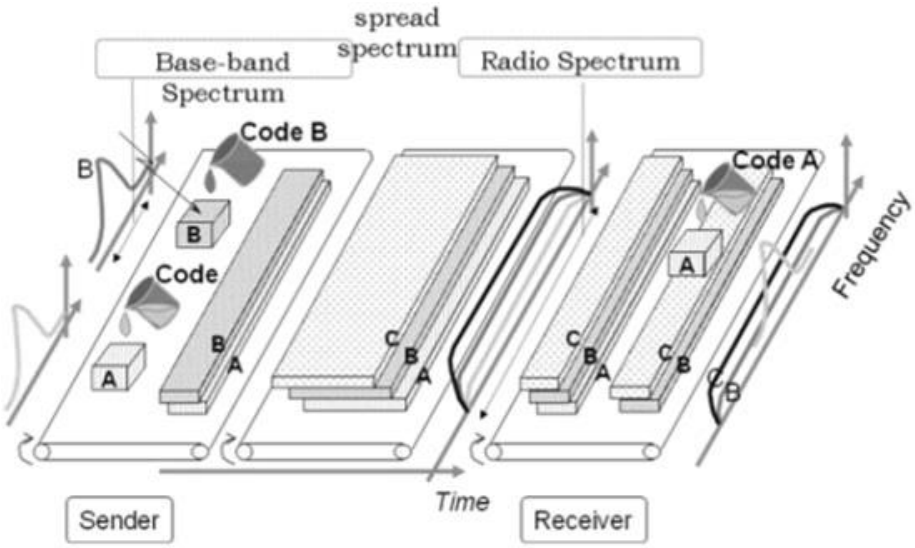
How Does CDMA Work?

CDMA allows up to 61 concurrent users in a 1.2288 MHz channel by processing each voice packet with two PN codes. There are 64 Walsh codes available to differentiate between calls

and theoretical limits. Operational limits and quality issues will reduce the maximum number of calls somewhat lower than this value.

In fact, many different "signals" baseband with different spreading codes can be modulated on the same carrier to allow many different users to be supported. Using different orthogonal codes, interference between the signals is minimal. Conversely, when signals are received from several mobile stations, the base station is capable of isolating each as they have different orthogonal spreading codes.

The following figure shows the technicality of the CDMA system. During the propagation, we mixed the signals of all users, but by that you use the same code as the code that was used at the time of sending the receiving side. You can take out only the signal of each user.



CDMA Capacity

The factors deciding the CDMA capacity are −

- Processing Gain
- Signal to Noise Ratio
- Voice Activity Factor
- Frequency Reuse Efficiency

Capacity in CDMA is soft, CDMA has all users on each frequency and users are separated by code. This means, CDMA operates in the presence of noise and interference.

In addition, neighboring cells use the same frequencies, which means no re-use. So, CDMA capacity calculations should be very simple. No code channel in a cell, multiplied by no cell. But it is not that simple. Although not available code channels are 64, it may not be possible to use a single time, since the CDMA frequency is the same.

Centralized Methods

- The band used in CDMA is 824 MHz to 894 MHz (50 MHz + 20 MHz separation).
- Frequency channel is divided into code channels.
- 1.25 MHz of FDMA channel is divided into 64 code channels.

Processing Gain

CDMA is a spread spectrum technique. Each data bit is spread by a code sequence. This means, energy per bit is also increased. This means that we get a gain of this.

P (gain) = 10log (W/R)

W is Spread Rate

R is Data Rate

For CDMA P (gain) = 10 log (1228800/9600) = 21dB

This is a gain factor and the actual data propagation rate. On an average, a typical transmission condition requires a signal to the noise ratio of 7 dB for the adequate quality of voice.

Translated into a ratio, signal must be five times stronger than noise.

Actual processing gain = P (gain) - SNR

= 21 – 7 = 14dB

CDMA uses variable rate coder

The Voice Activity Factor of 0.4 is considered = -4dB.

Hence, CDMA has 100% frequency reuse. Use of same frequency in surrounding cells causes some additional interference.

In CDMA frequency, reuse efficiency is 0.67 (70% eff.) = -1.73dB

Advantages of CDMA

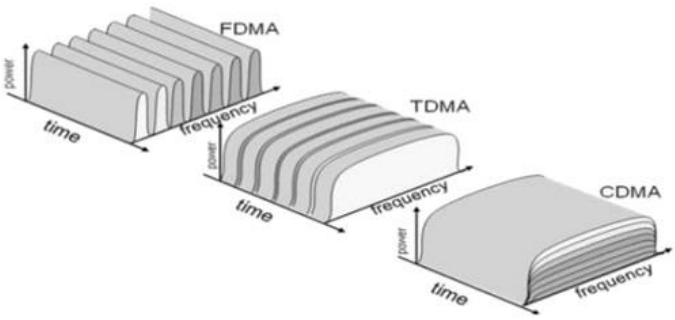
CDMA has a soft capacity. The greater the number of codes, the more the number of users. It has the following advantages −

- CDMA requires a tight power control, as it suffers from near-far effect. In other words, a user near the base station transmitting with the same power will drown the signal latter. All signals must have more or less equal power at the receiver
- Rake receivers can be used to improve signal reception. Delayed versions of time (a chip or later) of the signal (multipath signals) can be collected and used to make decisions at the bit level.
- Flexible transfer may be used. Mobile base stations can switch without changing operator. Two base stations receive mobile signal and the mobile receives signals from the two base stations.
- Transmission Burst − reduces interference.

Disadvantages of CDMA

The disadvantages of using CDMA are as follows −

- The code length must be carefully selected. A large code length can induce delay or may cause interference.
- Time synchronization is required.
- Gradual transfer increases the use of radio resources and may reduce capacity.
- As the sum of the power received and transmitted from a base station needs constant tight power control. This can result in several handovers.



CDMA Network is the system meant to regulate CDMA technology. It includes all aspects and functionality starting from the base station, transmitting antenna, receiving antenna, to mobile switching centers.

The MSC is connected to the outside world, i.e. the fixed line network. MSC can also be connected to several other MSCs.

CDMA Identities

Network Identities −

- SID (System Identity)
- NID (Network Identity)

3.5. Cellular Internet Access-An Overview of Cellular Network Architecture, 3g Cellular Data Networks

3.5.1. An overview of cellular network architecture

Cellular Network is formed of some cells, cell covers a geographical region, has a base station analogous to 802.11 AP which helps mobile users attach to the network and there is an air-interface of physical and data link layer protocol between mobile and base station. All these base stations are connected to Mobile Switching Center which connects cells to wide-area net, manages call setup, and handles mobility.

There is a certain radio spectrum that is allocated to the base station and to a particular region and that now needs to be shared. There are two techniques for sharing mobile-to-base station radio spectrum are:

1. **Combined FDMA/TDMA:** It divides the spectrum into frequency channels and divides each channel into time slots.
2. **Code Division Multiple Access (CDMA):** It allows the reuse of the same spectrum over all cells. Net capacity improvement. Two frequency bands are used one of which is for the forwarding channel (cell-site to subscriber) and one for the reverse channel (sub to cell-site).

Cell Fundamentals

In practice, cells are of arbitrary shape(close to a circle) because it has the same power on all sides and has same sensitivity on all sides, but putting up two-three circles together may result in interleaving gaps or may intersect each other so in order to solve this problem we can use equilateral triangle, square or a regular hexagon in which hexagonal cell is close to a circle used for a system design. Co-channel reuse ratio is given by:

DL/RL = Square root of (3N)

Where,

DL = Distance between co-channel cells

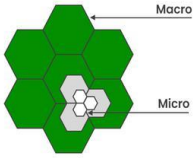
RL = Cell Radius

N = Cluster Size

The number of cells in a cluster N determines the amount of co-channel interference and also the number of frequency channels available per cell.

Cell Splitting

When the number of subscribers in a given area increases allocation of more channels covered by that channel is necessary, which is done by cell splitting. A single small cell midway between two co-channel cells is introduced.



Need for Cellular Hierarchy

Extending the coverage to the areas that are difficult to cover by a large cell. Increasing the capacity of the network for those areas that have a higher density of users. An increasing number of wireless devices and the communication between them.

Cellular Hierarchy

1. **Femtocells:** The smallest unit of the hierarchy, these cells need to cover only a few meters where all devices are in the physical range of the uses.
2. **Picocells:** The size of these networks is in the range of a few tens of meters, e.g., WLANs.
3. **Microcells:** Cover a range of hundreds of meters e.g. in urban areas to support PCS which is another kind of mobile technology.
4. **Macrocells:** Cover areas in the order of several kilometers, e.g., cover metropolitan areas.
5. **Mega cells:** Cover nationwide areas with ranges of hundreds of kilometers, e.g., used with satellites.

Fixed Channel Allocation

For a particular channel, the frequency band which is associated is fixed. The total number of channels is given by

$Nc = W/B$

Where,

W = Bandwidth of the available spectrum,

B = Bandwidth needed by each channels per cell,

Cc = Nc/N where N is the cluster size

Adjacent radio frequency bands are assigned to different cells. In analog, each channel corresponds to one user while in digital each RF channel carries several time slots or codes (TDMA/CDMA). Simple to implement as traffic is uniform.

3.5.2. Global System for Mobile (GSM) Communications

GSM uses 124 frequency channels, each of which uses an 8-slot Time Division Multiplexing (TDM) system. There is a frequency band that is also fixed. Transmitting and receiving do not happen in the same time slot because the GSM radios cannot transmit and receive at the same time and it takes time to switch from one to the other. A data frame is transmitted in 547 microseconds, but a transmitter is only allowed to send one data frame every 4.615 microseconds since it is sharing the channel with seven other stations. The gross rate of each channel is 270, 833 bps divided among eight users, which gives 33.854 kbps gross.

Control Channel (CC)

Apart from user channels, there are some control channels which is used to manage the system.

1. The broadcast control channel (BCC): It is a continuous stream of output from the base station's identity and the channel status. All mobile stations monitor their signal strength to see when they moved into a new cell.
2. The dedicated control channel (DCC): It is used for location updating, registration, and call setup. In particular, each base station maintains a database of mobile stations. Information is needed to maintain this database is sent on the dedicated control channel.

Common Control Channel

Three logical sub-channels are:

1. Is the paging channel, that the base station uses to announce incoming calls. Each mobile station monitors it continuously to watch for calls it should answer.
2. Is the random access channel that allows the users to request a slot on the dedicated control channel. If two requests collide, they are garbled and have to be retried later.
3. Is the access grant channel which is the announced assigned slot.

3.5.3. 3G Cellular Data Networks

G refers to the third generation of cellular technology that enables mobile telephony. The third-generation standard follows two earlier generations that were deployed on mobile networks and across mobile phones.

The International Telecommunication Union (ITU) defined the third generation of mobile telephony standards International Mobile Telecommunications 2000 (IMT-2000) to facilitate

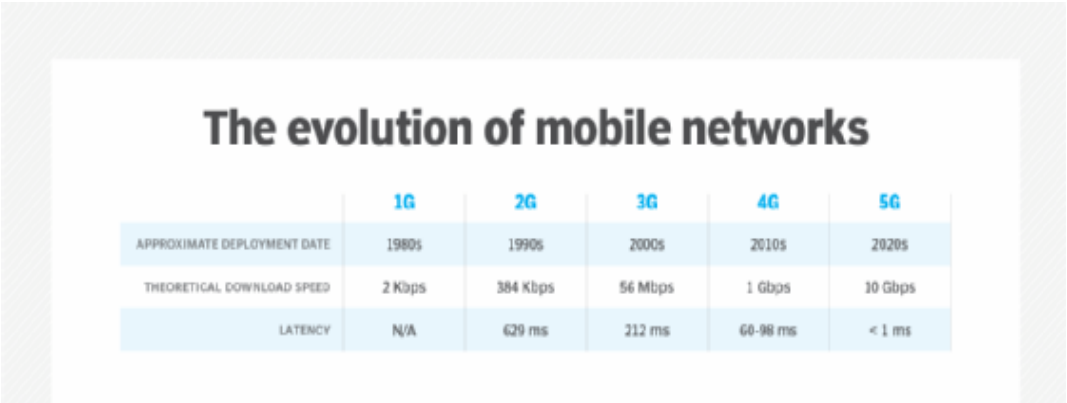
growth, increase bandwidth and support more diverse applications. For example, Global System for Mobile Communications (GSM) technologies could deliver not only voice, but also circuit-switched data across mobile phone networks at speeds up to 14.4 kilobits per second (Kbps). To support mobile multimedia applications, however, the 3G standard had to deliver packet-switched data with better spectral efficiency at far greater speeds.

History of mobile telephony

The first-generation (1G) standard got its start in the early '80s with commercial deployment of Advanced Mobile Phone Service (AMPS) cellular networks by network providers. Early AMPS networks used frequency division multiple access (FDMA) to carry analog voice over channels in the 800 megahertz (MHz) frequency band across radio access technology.

The second-generation (2G) standard emerged in the '90s when mobile operators deployed two competing digital voice standards. In North America, some operators adopted Interim Standard 95 (IS-95), which used code division multiple access (CDMA) technologies to multiplex up to 64 calls per channel in the 800 MHz band. Across the world, many operators adopted the GSM standard, which used time division multiple access (TDMA) to multiplex up to eight calls per channel in the 900 and 1,800 MHz bands.

To get from the 2G to the 3G standard, mobile operators had to upgrade existing networks while planning new mobile broadband networks. This led to the establishment of two distinct 3G families: 3GPP and 3GPP2.



The 3rd Generation Partnership Project (3GPP) was formed in 1998 to foster deployment of 3G networks that descended from GSM. 3GPP technologies evolved as follows:

- General Packet Radio Service (GPRS) offered speeds up to 114 Kbps.
- Enhanced Data Rates for Global Evolution (EDGE) reached up to 384 Kbps.
- Universal Mobile Telecommunications System (UMTS) Wideband CDMA (WCDMA) offered downlink speeds up to 1.92 megabits per second (Mbps).
- High-Speed Downlink Packet Access (HSDPA) boosted the downlink to 14 Mbps.
- Long-Term Evolution (LTE) -- or Evolved Universal Terrestrial Radio Access (E-UTRA) -- aimed for 100 Mbps.

GPRS deployments began in 2000, followed by EDGE in 2003. While these technologies are defined by IMT-2000, they are sometimes called *2.5G* because they did not offer multimegabit data rates. EDGE was superseded by HSDPA and its uplink partner, HSUPA. According to 3GPP, there were 166 HSDPA networks in 75 countries at the end of 2007. The next step for GSM operators, E-UTRA, was based on specifications completed in late 2008.

A second organization, 3rd Generation Partnership Project 2 (3GPP2), was formed to help North American and Asian operators using CDMA2000 transition to 3G. 3GPP2 technologies evolved as follows:

- Single-Carrier Radio Transmission Technology (1xRTT) offered speeds up to 144 Kbps.
- Evolution-Data Optimized (EV-DO) increased downlink speeds up to 2.4 Mbps.
- EV-DO Revision A (Rev. A) boosted downlink peak speed to 3.1 Mbps and reduced latency.
- EV-DO Revision B (Rev. B) can use two to 15 channels, with each downlink peaking at 4.9 Mbps.
- Ultra Mobile Broadband (UMB) reached 288 Mbps on the downlink.

1xRTT became available in 2002, followed by commercial EV-DO Release 0 in 2004. Here again, 1xRTT is referred to as *2.5G* because it served as a transitional step to EV-DO.

EV-DO standards were extended twice. Rev. A services emerged in 2006 and were succeeded by products that use Rev. B to increase data rates by transmitting over multiple channels. 3GPP2's next-generation technology, UMB, did not catch on, as many CDMA mobile phone network operators moved to LTE instead.

LTE evolved to become a core component of the fourth-generation (4G) wireless technology standard, providing a higher peak data transfer rate than 3G. The 4G standard can provide up to 100 Mbps downstream and up to 30 Mbps of upstream bandwidth capability. Mobile operators first deployed 4G in 2009.

4G has since been superseded by the fifth-generation (5G) wireless standard, which began deployment in 2019. 5G technologies offer even higher peak speeds than 4G, with a theoretical peak speed that can reach up to 20 gigabits per second (Gbps).

Areas of application

There are a number of use cases and applications for 3G technologies:

- **Mobile networks.** The 3G standard is most commonly used within mobile networks and their radio access technology platforms.
- **Mobile phones/smartphones.** The most common 3G devices are mobile phones for basic voice and text connectivity, as well as smartphones that provide more capabilities to users for data transfer and access. 3G-enabled smartphones helped drive development and adoption of mobile internet sites.
- **Mobile broadband routers.** For both consumer and business use cases, mobile broadband routers that support the 3G standard can enable internet connectivity for endpoint devices that are attached to the router -- either via a wire or Wi-Fi.
- **Computer modems.** 3G Universal Serial Bus (USB) modems for computer laptops enable cellular connectivity for devices that don't have access to Wi-Fi or a wired network connection.
- **Cellular backup.** In various devices -- including wired routers and home alarm systems -- a cellular backup option is sometimes available. Cellular backup enables the primary connectivity via a fixed wired connection, and in the event of a failure, the device will fail over to use the cellular connection.
- **Telematics.** 3G is widely deployed to enable telematics applications for the automotive industry, enabling vehicles to share information for transportation and fleet management use cases.

Differences among 3G, 4G and 5G

There are a number of key differences across the 3G, 4G and 5G standards. The most basic and primary difference is that each successive generation offers the promise of an increased peak speed.

However, speed isn't the only differentiator as access technology also changes in each platform, requiring mobile phone network providers and mobile phone users to upgrade devices and radio access technology.

- **3G** provides a peak speed of up to 14 Mbps and works at frequencies up to 2.1 gigahertz (GHz).
- **4G** has a peak speed of up to 100 Mbps and works at frequencies up to 2.5 GHz. 4G technology includes multiple input, multiple output (MIMO) and orthogonal frequency-division multiplexing (OFDM) technologies to increase speed and capacity over 3G.
- **5G** has a theoretical peak speed of 20 Gbps and works at frequencies of up to 95 GHz. 5G also offers the promise of reduced power consumption over its predecessor, which will enable more use cases, including embedded deployment for internet of things (IoT) devices.

2G and 3G cellular network sunset

While there has been a degree of backward compatibility across cellular network standards, enabling newer smartphones to run on mobile phone networks that support older standards, network providers around the world are moving to decommission the older 2G and 3G networks. The move is intended to help advance the adoption of newer standards and optimize radio access technology platform deployments that mobile phone network operators need to support.

3.6. 4G LTE Cellular networks - LTE Protocol Stacks -LTE Radio Access Network -Additional LTE functions

4G LTE networks are the next generation from the existing 3G networks - Universal Mobile Telecommunications System UMTS or 3rd generation.

LTE full form - What does LTE stands for?

LTE stands for Long Term Evolution.

Why?

It was indeed a "long-term evolution" for mobile networks (based on 3G tech at that time) when 4G came out in 2008.

The evolution took many years.

1G (First generation)	1980's	Let us talk to each other on mobile phones
2G	1990's	Let us send SMS and travel (roaming)
3G	2000's	Introduced some mobile internet experience
4G	2010's	Brought fast internet experience
5G	2020's	Brings super-fast Internet and much more

What does LTE mean on my phone? It means fast-internet connection. Mobile network generations: from 1G to 5G - Source Thales 2021

4G LTE is a global standard and success.

Who's in charge of defining and maintaining this technology?

The 3rd Generation Partnership Project (3GPP) standards group has developed and maintains LTE high-speed wireless technology.

3GGP unites telecom standard development organizations so that everybody uses the same technology.

And 4G LTE is a global success.

66% of mobile users have 4G LTE in Jan. 2022

LTE is a success worldwide, with 6.6 billion subscriptions connecting two-thirds of mobile users globally, according to the Global mobile Suppliers' Associations (GSA - January 2022 report).

- 791 telecom operators are running LTE networks in 240 countries and territories across the globe.
- 336 have rolled out LTE-Advanced networks.
- 227 have initiated VoLTE networks.

4G LTE is today the most current cell phone technology.

LTE vs 3G

Is 4G LTE the same as 3G?

No. LTE means 4G, and the visible difference is speed!

- 3G (third generation of mobile communications introduced in 2001) theoretically delivers 7.2Mbps and up to 3Mbps in practice.
- 3G HSPA+ (the advanced version - High-Speed Packet Access+) or 3G++ delivers up to 42Mps and up to 6Mbps in practice.

4G LTE data rates with 100Mbps are 2,5 times faster than 3G HSPA+ and 15 times faster than 3G.

LTE vs 4G: who's better?

What's the difference between LTE and 4G?

Well, they are not competing. LTE is the technology behind 4G (the fourth generation of mobile communications - an architecture).

All 4G phones utilize LTE technology in 2022. It brings high speed to mobile and broadband data.

How fast is 4G LTE?

In theory, LTE's maximum speed is 100Mbps. In practice, it tops at 15Mbps. Of course, it all depends on where you're located.

But what is LTE-A?

LTE-A, LTE-Advanced, 4G+, and LTE+

LTE-A, LTE-Advanced, 4G+, and LTE+ are all acronyms for the same 4G service. It's a faster version of LTE.

How fast is LTE-A? Again, in theory, LTE-A data rates are up to 300Mbps. In practice, you can expect between 40 to 90Mbps.

LTE-A is three times faster than LTE.

4G LTE features at a glance

LTE – often called 4th generation LTE or 4G LTE – have an all-IP flat networking structure. LTE is used for mobile, fixed, and portable broadband access.

LTE is designed for lower latency (the time it takes for data to travel in the network) and increased bandwidth - very interesting for the Internet of Things.

In fact, bandwidth increases can be as high as 100 Mbps on the downlink and up to 50 Mbps on the uplink.

- The higher bandwidth enables faster access to content and applications, particularly video applications that can only be offered today on fixed systems.
- The low latency enables time-sensitive applications like voice services.
- The all-IP architecture enables new converging services based on the IP Multimedia Subsystem (IMS).

There's more.

Voice over LTE (VoLTE)

What is VoLTE on your phone?

VoLTE means voice-over LTE. It's an improved version of 4G LTE for voice and video calls.

In essence, you get HD voice and video calls, and it's a great experience overall with better coverage and battery life increase.

If you upgrade to 4G LTE, make sure you also get 4G VoLTE.

4G LTE and India: massive expansion in 2020

Around 100m new 4G users were added in 2020, and 4G data consumption alone contributed to 99% of the total traffic in the country with a +60% annual growth, according to Times of India. Data traffic grew by x60 from 2015 to 2020. Early 2021, India is now a full 4G market with 99% of the 700m mobile internet subscribers using 4G LTE for going online.

But 4G LTE is also decisive for IoT devices (connected things).

4G LTE for the IoT (Internet of Things)

There are three broad categories within 4G – which is mainstream to connect industry-grade IoT devices today.

- The LPWAN, low powered wide area network, has two variants, which are category M (Machine to Machine) (Cat-M or LTE-M) and category NB-IoT (Cat NB-IoT).
- The mid-range bandwidth is category LTE-1 (LTE Cat 1).
- The high bandwidth applications typically use networks called LTE Advanced (LTE-A) or LTE Advanced Pro.

4G comes in different flavors for connected objects. Here are some examples of IoT devices and their specific bandwidth needs. The suitable 4G LTE module can then be selected according to the need.

LTE back-ward compatibility

LTE is compatible with existing mobile networks to provide maximum flexibility.

This feature encompasses formerly deployed network technology, including GSM, cdmaOne, W-CDMA (UMTS), and CDMA2000.

Wi-Fi vs LTE

Wi-Fi was created in 2000 by the Wi-Fi Alliance. It is a set of wireless local (limited range) area network (LAN) protocols for short-distance connectivity.

Access to the Internet is most commonly done via a wireless router.

Wi-Fi is based on IEEE 802.11.

LTE is a broadband (unlimited range) wireless network technology.

First, Wi-Fi-certified products came with IEEE 802.11b (yes, b came before a).

How fast is Wi-Fi?

The theoretical speed for version b is 11Mbps and 5.5Mps actual. Version a offers 54Mbps in theory and about 20Mps in reality. The maximum range is 10m for version b and 140m for version a.

Wi-Fi comes in many flavors now, and speed performance continues to improve with each new generation.

Before wireless – If you're old enough to remember – network connections on a LAN or over the Internet were done via Ethernet wired connections or a modem over phone lines.

Wi-Fi works best for line-of-sight use, and walls, pillars, or TV screens (obstacles) may seriously reduce range. Practical performance is also related to distance from the router.

Wi-Fi speed continues to improve with each new generation.

Is 5G the same as 4G LTE?

No, it's not. LTE is a 4G technology.

5G is the fifth generation of mobile network technology. 5G does not replace 4G LTE. 4G and 5G work together.

5G network technology, 5G services, and 5G-enabled smartphones have started their roll-out.

Needless to say, there's no such thing as 5G LTE.

4G - 5G milestones

- 4G (2008) networks brought all-IP services (Voice and Data), a fast internet experience, with unified network architectures and protocols.
- 4 G LTE (for Long Term Evolution), starting in 2009, is doubling data speeds.

- 5G (starting in 2019) networks expand broadband wireless services beyond mobile Internet to IoT and critical communications segments with speed up to X10 compared to 4G LTE.



When will 5G take over?

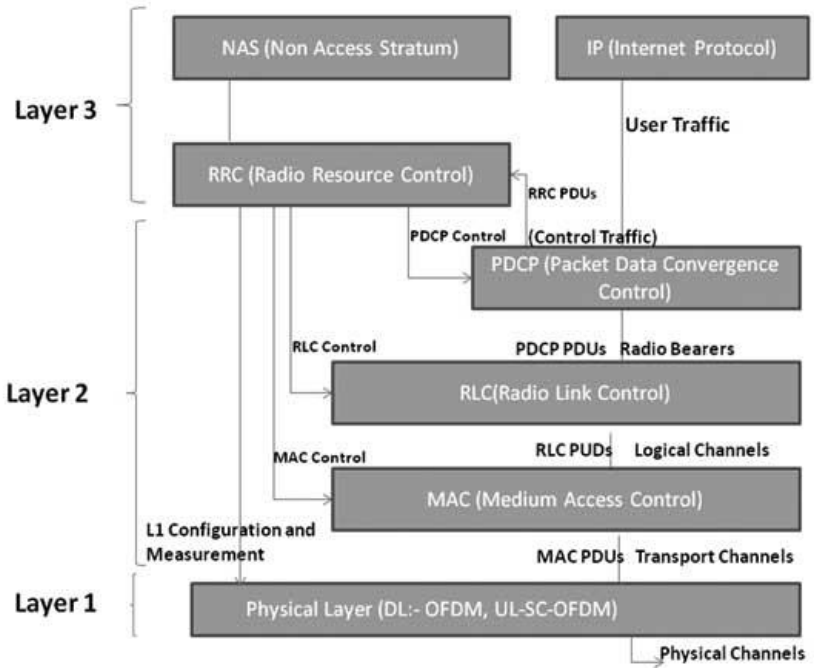
In short, 2019 was when 5G lifts off, and 2020 onward is when 5G networks and 5G phones will be available *en masse*.

ABI Research sees 5G overtake 4G by 2025.

5G with 10Gbps is up to 100 times faster than 4G LTE.

3.6.1. LTE Protocol Stacks

Let's have a close look at all the layers available in E-UTRAN Protocol Stack which we have seen in previous chapter. Below is a more ellaborated diagram of E-UTRAN Protocol Stack:



Physical Layer (Layer 1)

Physical Layer carries all information from the MAC transport channels over the air interface. Takes care of the link adaptation (AMC), power control, cell search (for initial synchronization and handover purposes) and other measurements (inside the LTE system and between systems) for the RRC layer.

Medium Access Layer (MAC)

MAC layer is responsible for Mapping between logical channels and transport channels, Multiplexing of MAC SDUs from one or different logical channels onto transport blocks (TB) to be delivered to the physical layer on transport channels, de multiplexing of MAC SDUs from one or different logical channels from transport blocks (TB) delivered from the physical layer on transport channels, Scheduling information reporting, Error correction through HARQ, Priority handling between UEs by means of dynamic scheduling, Priority handling between logical channels of one UE, Logical Channel prioritization.

Radio Link Control (RLC)

RLC operates in 3 modes of operation: Transparent Mode (TM), Unacknowledged Mode (UM), and Acknowledged Mode (AM).

RLC Layer is responsible for transfer of upper layer PDUs, error correction through ARQ (Only for AM data transfer), Concatenation, segmentation and reassembly of RLC SDUs (Only for UM and AM data transfer).

RLC is also responsible for re-segmentation of RLC data PDUs (Only for AM data transfer), reordering of RLC data PDUs (Only for UM and AM data transfer), duplicate detection (Only for UM and AM data transfer), RLC SDU discard (Only for UM and AM data transfer), RLC re-establishment, and protocol error detection (Only for AM data transfer).

Radio Resource Control (RRC)

The main services and functions of the RRC sublayer include broadcast of System Information related to the non-access stratum (NAS), broadcast of System Information related to the access stratum (AS), Paging, establishment, maintenance and release of an RRC connection between the UE and E-UTRAN, Security functions including key management, establishment, configuration, maintenance and release of point to point Radio Bearers.

Packet Data Convergence Control (PDCP)

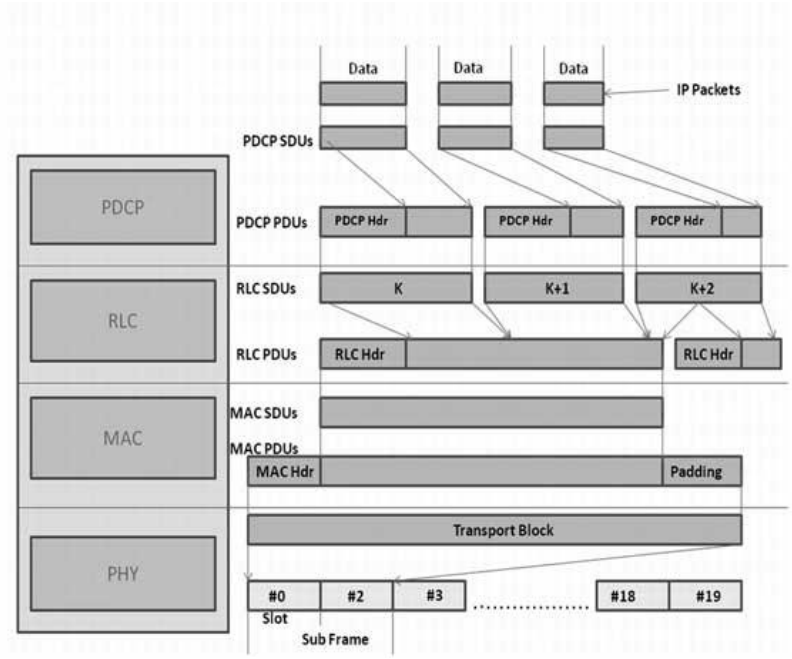
PDCP Layer is responsible for Header compression and decompression of IP data, Transfer of data (user plane or control plane), Maintenance of PDCP Sequence Numbers (SNs), In-sequence delivery of upper layer PDUs at re-establishment of lower layers, Duplicate elimination of lower layer SDUs at re-establishment of lower layers for radio bearers mapped on RLC AM, Ciphering and deciphering of user plane data and control plane data, Integrity protection and integrity verification of control plane data, Timer based discard, duplicate discarding, PDCP is used for SRBs and DRBs mapped on DCCH and DTCH type of logical channels.

Non Access Stratum (NAS) Protocols

The non-access stratum (NAS) protocols form the highest stratum of the control plane between the user equipment (UE) and MME.

NAS protocols support the mobility of the UE an d the session management procedures to establish and maintain IP connectivity between the UE and a PDN GW.

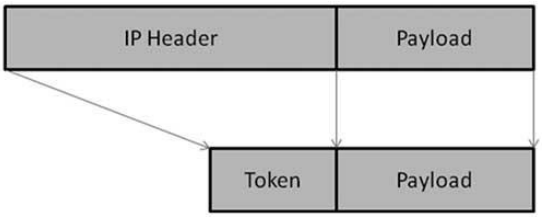
Below is a logical digram of E-UTRAN Protocol layers with a depiction of data flow through various layers:



Packets received by a layer are called Service Data Unit (SDU) while the packet output of a layer is referred to by Protocol Data Unit (PDU). Let's see the flow of data from top to bottom:

- IP Layer submits PDCP SDUs (IP Packets) to the PDCP layer. PDCP layer does header compression and adds PDCP header to these PDCP SDUs. PDCP Layer submits PDCP PDUs (RLC SDUs) to RLC layer.

PDCP Header Compression : PDCP removes IP header (Minimum 20 bytes) from PDU, and adds Token of 1-4 bytes. Which provides a tremendous savings in the amount of header that would otherwise have to go over the air.



- RLC layer does segmentation of these SDUS to make the RLC PDUs. RLC adds header based on RLC mode of operation. RLC submits these RLC PDUs (MAC SDUs) to the MAC layer.

RLC Segmentation : If an RLC SDU is large, or the available radio data rate is low (resulting in small transport blocks), the RLC SDU may be split among several RLC PDUs. If the RLC SDU is small, or the available radio data rate is high, several RLC SDUs may be packed into a single PDU.

- MAC layer adds header and does padding to fit this MAC SDU in TTI. MAC layer submits MAC PDU to physical layer for transmitting it onto physical channels.
- Physical channel transmits this data into slots of sub frame.

The information flows between the different protocols are known as channels and signals. LTE uses several different types of logical, transport and physical channel, which are distinguished by the kind of information they carry and by the way in which the information is processed.

- **Logical Channels** : Define **whattype** of information is transmitted over the air, e.g. traffic channels, control channels, system broadcast, etc. Data and signalling messages are carried on logical channels between the RLC and MAC protocols.
- **Transport Channels** : Define **howis** something transmitted over the air, e.g. what are encoding, interleaving options used to transmit data. Data and signalling messages are carried on transport channels between the MAC and the physical layer.
- **Physical Channels** : Define **whereis** something transmitted over the air, e.g. first N symbols in the DL frame. Data and signalling messages are carried on physical channels between the different levels of the physical layer.

3.6.2. LTE Radio Access Network

A Radio Access Network (RAN) is the part of a telecommunications system that connects individual devices to other parts of a network through radio connections. A RAN resides between user equipment, such as a mobile phone, a computer or any remotely controlled machine, and provides the connection with its core network. The RAN is a major component of wireless telecommunications and has evolved through the generations of mobile networking leading up to 5G.



CableFree 4G LTE Radio Access Network. CableFree offers LTE RAN with 4G and 5G

A RAN provides access and coordinates the management of resources across the radio sites. A handset or other device is wirelessly connected to a backbone, or core network, and the RAN sends its signal to various wireless end points, so it can travel with other networks' traffic. A single handset/phone could be connected at the same time to multiple RANs, which is sometimes called dual-mode handsets.

The term radio access network (RAN) has been in use since the beginning of cellular technology and has evolved through the generations of mobile communications (from 1G up to 5G today). Components of the RAN include a base station and antennas that cover a given region depending on their capacity, plus required core network items.

What components are used to make up a Radio Access Network?

RAN components include a base station and antennas that cover a specific region depending on their capacity, as well as core network including a RAN Controller. In modern LTE networks an EPC is required at the core.

Recent evolution of RAN architecture divides the user plane from the control plane into separate elements. User data messages can then be exchanged by the RAN controller through one software-defined networking (SDN) switch and a second set through a control-based interface. This separation allows the RAN to be more flexible, accommodating for network

functions virtualization (NFV) techniques such as network slicing and high MIMO that are necessary for 5G

Examples of radio access network types are:

- GRAN: GSM radio access network
- GERAN: essentially the same as GRAN but specifying the inclusion of EDGE packet radio services
- UTRAN: UMTS radio access network
- E-UTRAN: The Long Term Evolution (LTE) high speed and low latency radio access network

RAN Base Stations

In modern 4G networks the Base Station is termed eNodeB (Evolved NodeB). For 5G networks, the term gNodeB (Next Generation NodeB) is used. The base station takes digital packets from the network core (typically the EPC) and synthesises the radio signals for transmission. Modern eNodeB and gNodeB base stations typically use Software Defined Radio (SDR) for this purpose.

The Base Station may comprise of 1 sector (an example being a Small Cell) up to 3 sectors or more (typically a Macro Site) for longer range and higher capacity. A modern multi-band LTE Base station may transmit on more than one carrier, with the ability of Carrier Aggregation across multiple bands to provide higher user throughput.

BBU, RRH & Antennas

The Base Station may be split into a Baseband Unit (BBU) plus one or more Remote Radio Heads (RRH) which is a typical design for 4G LTE, or may be combined into a single unit for a Small Cell featuring 1 or 2 sectors only. The Radio Heads are typically mounted up on the tower next to the Sector Antenna(s) with short RF jumper cables for low signal loss to ensure maximal coverage.

The Remote Radio Heads typically feature MIMO connections to the antennas with 2×2, 4×4, 8×8 or even higher count for Massive MIMO. Generally, high order MIMO only works effectively at shorter distances in urban environments. However, 8×8 configurations can also

feature beamsteering to enhance range and capacity at the cell edge, making 8×8 MIMO potentially useful for long range & rural broadband applications.

Antennas used for the RAN may vary from an Omnidirectional or patch antenna for single-sector short-range coverage up to a cluster of Sector Antennas which may be designed with 3, 4, 6 or more sectors to create full 360-degree coverage from the base site to users. Generally, higher gain antennas are used for long range applications, whereas a short-range Omnidirectional or patch antenna has lower gain

The RAN Controller

The RAN controller controls the radio nodes that are connected to it. The network controller performs radio resource management, mobility management, and data encryption. It connects to the circuit-switched core network and the packet-switched core network, depending on the type of RAN.

In a modern 4G network an Evolved Packet Core (EPC) is central to the RAN core network. The EPC. For 5G, a Next Generation Core (NG, NextGen core) may be required. The requirement for an EPC makes LTE more complex than simpler wireless technologies such as WiFi, but brings many benefits such as seamless roaming and advanced user management. Some vendors such as CableFree offer the option to virtualise an embedded EPC within an eNodeB base station, enabling simpler LTE implementations with fewer boxes, reduced complexity and cost.

Modern RAN architectures separate the user plane from the control plane into different network elements. In this scenario, the RAN controller can exchange user data messages through one software-defined networking (SDN) switch, and a second set with base stations via a second control-based interface. This separation of the control plane and data plane will be an essential aspect of the flexible 5G RAN, as it aligns with SDN and network functions virtualization (NFV) techniques such as service chaining and network slicing.

Evolution of the RAN for 5G and beyond

Radio access networks have evolved significantly since their origins to the point at which today's RANs support multiple-input, multiple-output (MIMO) antennas, wider spectrum bandwidths, multi-band carrier aggregation, and so on — with many increases and enhancements for today with 5G and for the future.

3.7. 5G Cellular Networks

In telecommunications, **5G** is the fifth-generation technology standard for cellular networks, which cellular phone companies began deploying worldwide in 2019, and is the successor to 4G technology that provides connectivity to most current mobile phones.

Like its predecessors, 5G networks are cellular networks, in which the service area is divided into small geographical areas called *cells*. All 5G wireless devices in a cell are connected to the Internet and the telephone network by radio waves through a base station and antennae in the cell. The new networks have higher download speeds, with a peak speed of 10 gigabits per second (Gbit/s) when there is only one user in the network. 5G has higher bandwidth to deliver faster speeds than 4G and can connect more devices, improving the quality of Internet services in crowded areas.[2] Due to the increased bandwidth, it is expected the 5G networks will increasingly be used as general internet service providers (ISPs), competing with existing ISPs such as cable internet, and also will make possible new applications in internet-of-things (IoT) and machine-to-machine areas. Cell phones with 4G capability alone are not able to use the 5G networks.

Performance

1.Speed

5G is capable of delivering significantly faster data rates than 4G, with peak data rates of up to 20 gigabits per second (Gbps). Furthermore, average 5G download speeds have been recorded at 186.3 Mbps in the U.S. by T-Mobile, while South Korea leads globally with average speeds of 432 megabits per second (Mbps). 5G networks are also designed to provide significantly more capacity than 4G networks, with a projected 100-fold increase in network capacity and efficiency.

The most widely used form of 5G, sub-6 GHz 5G (mid-band), is capable of delivering data rates ranging from 10 to 1,000 megabits per second (Mbps), with a much greater reach than mmWave bands.

2.Latency

In 5G, the ideal "air latency" is of the order of 8 to 12 milliseconds i.e., excluding delays due to HARQ retransmissions, handovers, etc. Retransmission latency and backhaul latency to the server must be added to the "air latency" for correct comparisons. Verizon reported the latency

on its 5G early deployment is 30 ms Edge Servers close to the towers can probably reduce latency to between 10 and 15 milliseconds

Latency is much higher during handovers; ranging from 50 to 500 milliseconds depending on the type of handover. Reducing handover interruption time is an ongoing area of research and development; options include modifying the handover margin (offset) and the time-to-trigger (TTT).

3.Error rate

5G uses adaptive modulation and coding scheme (MCS) to keep the block error rate (BLER) extremely low. Whenever the error rate crosses a (very low) threshold the transmitter will switch to a lower MCS, which will be less error-prone. This way speed is sacrificed to ensure an almost zero error rate.

4.Range

The range of 5G depends on many factors: transmit power, frequency, and interference. For example, mmWave (e.g.:band n258) will have a lower range than mid-band (e.g.: band n78) which will have a lower range than low-band (e.g.: band n5)

Given the marketing hype on what 5G can offer, simulators and drive tests are used by cellular service providers for the precise measurement of 5G performance.

Standards

Initially, the term was associated with the International Telecommunication Union's IMT-2020 standard, which required a theoretical peak download speed of 20 gigabits per second and 10 gigabits per second upload speed, along with other requirements. Then, the industry standards group 3GPP chose the 5G NR (New Radio) standard together with LTE as their proposal for submission to the IMT-2020 standard.

5G NR can include lower frequencies (FR1), below 6 GHz, and higher frequencies (FR2), above 24 GHz. However, the speed and latency in early FR1 deployments, using 5G NR software on 4G hardware (non-standalone), are only slightly better than new 4G systems, estimated at 15 to 50% better.

5G NR

5G NR (New Radio) is the de-facto air interface developed for 5G networks. It is the global standard for 3GPP 5G networks.

The study of NR within 3GPP started in 2015, and the first specification was made available by the end of 2017. While the 3GPP standardization process was ongoing, the industry had already begun efforts to implement infrastructure compliant with the draft standard, with the first large-scale commercial launch of 5G NR having occurred in the end of 2018. Since 2019, many operators have deployed 5G NR networks and handset manufacturers have developed 5G NR enabled handsets.

5Gi

5Gi is an alternative 5G variant developed in India. It was developed in a joint collaboration between IIT Madras, IIT Hyderabad, TSDSI, and the Centre of Excellence in Wireless Technology (CEWiT) 5Gi is designed to improve 5G coverage in rural and remote areas over varying geographical terrains. 5Gi uses Low Mobility Large Cell (LMLC) to extend 5G connectivity and the range of a base station.

In April 2022, 5Gi was merged with the global 5G NR standard in the 3GPP Release 17 specifications.

Pre-standard implementations

- 5G TF: American carrier Verizon used a pre-standard variation of 5G known as 5G TF (Verizon 5G Technical Forum) for Fixed Wireless Access in 2018. The 5G service provided to customers in this standard is incompatible with 5G NR. Verizon has since migrated to 5G NR.
- 5G-SIG: KT Corporation had a pre-standard varation of 5G developed called 5G-SIG. This was deployed at the Pyeongchang 2018 Winter Olympics.

Internet of things

In the Internet of things (IoT), 3GPP is going to submit evolution of NB-IoT and eMTC (LTE-M) as 5G technologies for the LPWA (Low Power Wide Area) use case.

Non-Terrestrial Network

Standards are being developed by 3GPP to provide access to end devices via non-terrestrial networks (NTN), ie. satellite or airborne signal relays to allow for better coverage outside of populated or otherwise hard to reach locations.

Several manufacturers have announced and released hardware that integrates 5G with satellite networks:

- Samsung Electronics introduced a standardized 5G NTN modem technology in Korea in February 2023, simulated on their Exynos Modem 5300, facilitating smartphone-satellite communication.
- MediaTek launched the world's first commercially available 5G IoT-NTN chipset, MT6825, capable of automatic satellite message receipt and extensive power efficiency.
- Qualcomm, in collaboration with Skylo, announced new satellite IoT solutions on June 22, 2023, including the Qualcomm 212S and 9205S modems, supporting the Qualcomm Aware platform for real-time asset tracking and device management.
- Motorola's Defy Satellite Link hotspot, powered by MediaTek's MT6825, became available in June 2023, providing a portable satellite messaging solution with robust battery life and built-in GPS.
- Rakuten Symphony, in collaboration with Supermicro, announced high-performing Open RAN technologies and storage systems for operators of cloud-based mobile services.

3.8. Managing Mobility In Cellular Networks, Wireless And Mobility-Impact On Higher Level Protocols

3.8.1.Managing Mobility In Cellular Networks

The mobile station attempts to find a suitable cell by passing through the list in descending order of received signal strength, the first BCCH channel, which satisfies a set of requirements it has selected.

Cell Selection Criteria

The requirements that a cell must satisfy before a mobile station can receive service from it are

- It should be a cell of the selected PLMN. The mobile station checks whether the cell is part of the selected PLMN.
- It should not be "barred". The PLMN operator may decide not to allow mobile stations to access certain cells. These cells may, for example only be used for handover traffic. Barred cell information is broadcast on the BCCH to instruct mobile stations not to access these cells.
- The radio path loss between the mobile station and the selected BTS must be above a threshold set by the PLMN operator.
- If no suitable cell is found then the MS enters a "limited service" state in which it can only make emergency calls.

Call to an Active Mobile Station

As an active mobile station (MS) moves in the coverage area of a public land mobile network (PLMN), it reports its movements so that it can be located as needed, using the update procedure locations. When a mobile services switching center (MSC) in the network needs to establish a call to a mobile station operating in its flow area, following things occur −

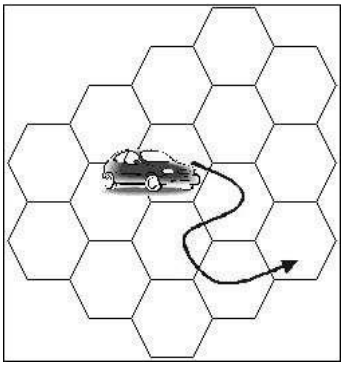
- A page message its broadcast which contains the identification code of the MS. Not every Base Station Controller (BSC) in the network is requested to transmit the page message. The broadcast is limited to a cluster of radio cells that together form a location area. The last reported position of the MS identifies the location area to be used for the broadcast.

- The MS monitors the page message transmitted by the radio cell in which it is located and, on detecting its own identification code, responds by transmitting a page response message to the Base Transceiver Station (BTS).
- Communication is then established between the MSC and the MS via BTS that received the page response message.

Location Update

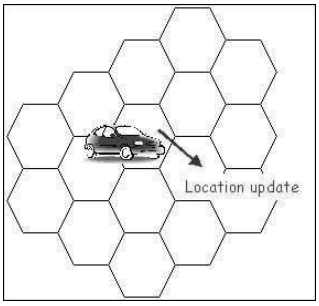
Case 1 − Location never updates.

If location never updates the implementation for location update, cost becomes zero. But we have to page every cell for locating the MS and this procedure will not be cost effective.

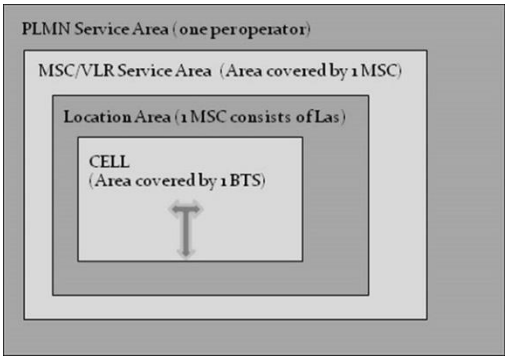


Case 2 − Location update is implemented.

Location updates are taking place as per the requirements of the network, may be time or movement or distance based. This procedure involves high cost, but we have to page single cell or few cells only for locating the MS and this procedure will be cost effective.



Network Configuration



The configuration of a Public Land Mobile Network (PLMN) is designed so that active mobile station moving in the network area is still able to report its position. A network consists of different areas −

- PLMN area
- Location area
- MSC area
- PLMN Area

A PLMN area is the geographical area in which land mobile communication services are provided to the public by a particular PLMN operator. From any position within a PLMN area, the mobile user can set up calls to another user of the same network, or to a user of another network. The other network may be a fixed network, another GSM PLMN, or another type of PLMN. Users of the same PLMN or users of other networks can also call a mobile user who is active in the PLMN area. When there are several PLMN operators, the geographical areas covered by their networks may overlap. The extent of a PLMN area is normally limited by national borders.

Location Area

To eliminate the need for network-wide paging broadcasts, the PLMN needs to know the approximate positions of the MSs that are active within its coverage area. To enable the approximate positions of any MS to be represented by a single parameter, the total area covered by the network is divided into location areas. A Location Area (LA) is a group of one or more radio cells. This group fulfills the following requirements −

- BTSs in one location area may be controlled by one or more BSCs.

- BSCs those serve the same location area are always connected to the same MSC.
- Radio cells with BTSs controlled by a common BSC can lie in different location areas.

Location Area Identity

Every radio transmitter in the PLMN broadcast, via a control channel BCCH, a Location Area Identity (LAI), code to identify the location area that it serves. When an MS is not engaged in a call, it automatically scans the BCCH transmitted by the base stations in the locality and selects the channel that is delivering the strongest signal. The LAI code broadcast by the selected channel identifies the location area in which the MS is currently situated. This LAI code is stored in the Subscriber Identity Module (SIM) of the mobile equipment.

As the MS moves through the network area, the signal received from the selected control channel gradually diminishes in strength until it is no longer the strongest. At this point the MS re-tunes to the channel that has become dominant and examines the LAI code that it is broadcasting. If the received LAI code differs from that stored on the SIM, then the MS has entered another location area and initiates a location update procedure to report the change to the MSC. At the end of the procedure, the LAI code in the SIM is also updated.

Location Area Identity Format

It is a Location Area Identity (LAI) code to identify the location area in a PLMN. The LAI code has three components −

Mobile Country Code (MCC)

The MCC is a 3-digit code that uniquely identifies the country of domicile of the mobile subscriber (for example, India 404). It is assigned by the ITU-T.

Mobile Network Code (MNC)

The MNC is a 2-digit code (3-digit code for GSM-1900) that identifies the home GSM PLMN of the mobile subscriber. If more than one GSM PLMN exists in a country, a unique MNC is assigned to each of them. It is assigned by the government of each country. (For example Cell one, Chennai 64).

Location Area Code (LAC)

The LAC component identifies a location area within a PLMN; it has a fixed length of 2 octets and can be coded using hexadecimal representation. It is assigned by an operator.

MSC areas

An MSC area is a region of the network in which GSM operations are controlled by a single MSC. An MSC area consists of one more location areas. The boundary of an MSC area follows the external boundaries of the location areas on its periphery. Consequently, a location area never spans beyond the boundary of an MSC area.

VLR area

A VLR area is region of the network that is supervised by a single Visitor Location Register (VLR). In theory, a VLR area may consist of one more MSC areas. In practice, however the functions of the VLR are always integrated with those of the MSC so that the terms "VLR area" and "MSC area" have become synonymous.

Location Related Databases

Two databases are used by Location Management to store MS location related data.

- Visitor Location Register(VLR)
- Home Location Register(HLR)

Visitor Location Register

A VLR contains a data record for each of the MS that are currently operating in its area. Each record contains a set of subscriber identity codes, related subscription information, and a Location Area Identity (LAI) code. This information is used by the MSC when handling calls to or from an MS in the area. When an MS moves from one area to another, the responsibility for its supervision passes from one VLR to another. A new data record is created by the VLR that has adopted the MS, and the old record is deleted. Provided that aninter-working agreement exists between the network operators concerned, data transaction can cross both network and national boundaries.

Home Location Register

The HLR contains information relevant to mobile subscribers who are fee-paying customers of the organization that operates the PLMN.

The HLR stores two types of information −

Subscription Information

The subscription information includes the IMSI and directory number allocated to the subscriber, the type of services provided and any related restrictions.

Location Information

The location information includes the address of the VLR in the area where the subscribers MS is currently located and the address of the associated MSC.

The location information enables incoming calls to be routed to the MS. The absence of this information indicates that the MS is inactive and cannot be reached.

When an MS moves from one VLR area to another, the location information in the HLR is updated with the new entry for the MS, using subscription data copied from the HLR. Provided that an inter-working agreement exists between the network operators, concerned data transactions can move across both network and national boundaries.

Types of Identification Numbers

During the performance of the location update procedure and the processing of a mobile call different types of numbers are used −

- Mobile Station ISDN Number(MSISDN)
- Mobile Subscriber Roaming Number(MSRN)
- International Mobile Subscriber Identity(IMSI)
- Temporary Mobile Subscriber Identity(TMSI)
- Local Mobile Station Identity(LMSI)

Each number is stored in the HLR and/or VLR.

Mobile Station ISDN Number

The MSISDN is the directory number allocated to the mobile subscriber. It is dialed to make a telephone call to the mobile subscriber. The number consists of Country Code (CC) of the country in which the mobile station is registered (e.g. India 91), followed by national mobile number which consists of Network Destination Code (NDC) and Subscriber Number (SN). An NDC is allocated to each GSM PLMN.

The composition of the MSISDN is such that it can be used as a global title address in the Signaling Connection Control Part (SCCP) for routing message to the HLR of the mobile subscriber.

Mobile Station Roaming Number

The MSRN is the number required by the gateway MSC to route an incoming call to an MS that is not currently under the control of the gateway MSISDN. Using a mobile, terminated call is routed to the MSC gateway. Based on this, MSISDN gateway MSC requests for a MSRN to route the call to the current visited MSC International Mobile Subscriber Identity (IMSI).

An MS is identified by its IMSI. The IMSI is embedded in the SIM of the mobile equipment. It is provided by the MS anytime it accesses the network.

Mobile Country Code (MCC)

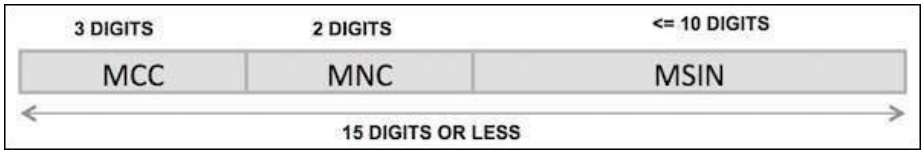
The MCC component of the IMSI is a 3-digit code that uniquely identifies the country of the domicile of the subscriber. It is assigned by the ITU-T.

Mobile Network Code (MNC)

The MNC component is a 2-digit code that identifies the home GSM PLMN of the mobile subscriber. It is assigned by the government of each country. For GSM-1900 a 3-digit MNC is used.

Mobile Subscriber Identification Number (MSIN)

The MSIN is a code that identifies the subscriber within a GSM PLMN. It is assigned by the operator.



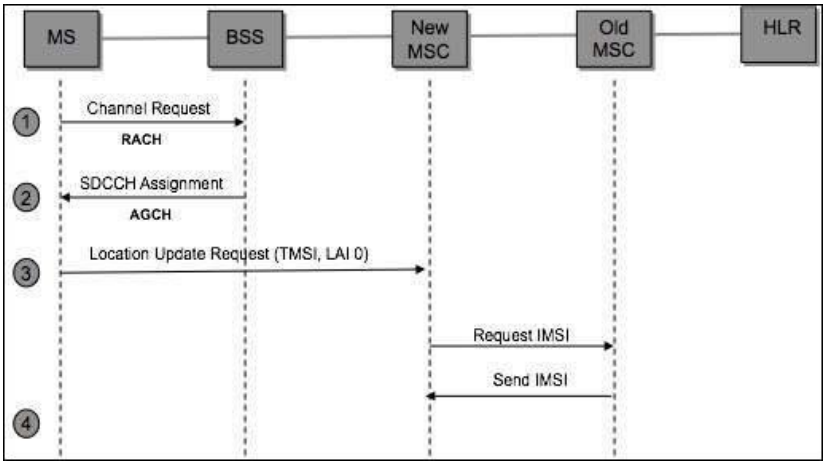
Temporary Mobile Subscriber Identity (TMSI)

The TMSI is an identity alias which is used instead of the IMSI when possible. The use of a TMSI ensures that the true identity of the mobile subscriber remains confidential by eliminating the need to transfer a non ciphered IMSI code over a radio link.

A VLR allocates a unique TMSI code to each mobile subscriber that is operating in its area. This code which is only valid within the area supervised by the VLR is used to identify the subscriber, in messages to and from the MS. When a change of location area also involves a change of VLR area, a new TMSI code is allocated and communicated to the MS. The MS stores the TMSI on its SIM. The TMSI consists of four octets.

Location Update Scenario

In the following location update scenario, it is assumed that an MS enters a new location area that is under control of a different VLR (referred to as the "new VLR") than the one where the MS is currently registered (referred to as the "old VLR"). The following diagram shows the steps of the mobile location update scenario.

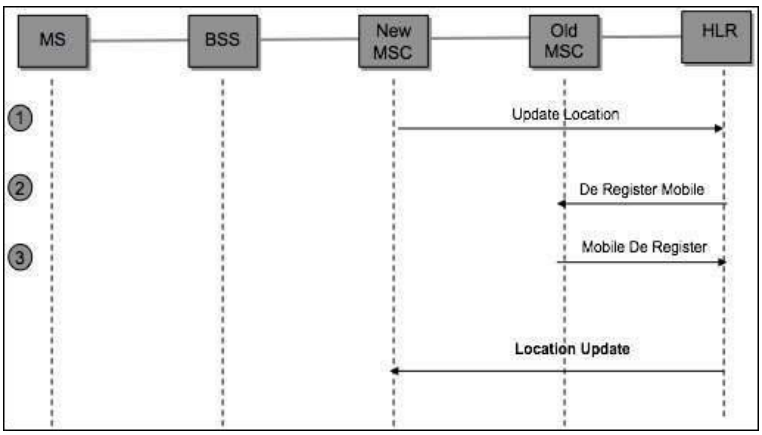


The MS enters a new cell area, listens to the Location Area Identity (LAI) being transmitted on the broadcast channel (BCCH), and compares this LAI with the last LAI (stored in the SIM) representing the last area where the mobile was registered.

- The MS detects that it has entered a new Location Area and transmits a Channel Request message over the Random Access Channel (RACH).
- Once the BSS receives the Channel Request message, it allocates a Stand-alone Dedicated Control Channel (SDCCH) and forwards this channel assignment information to the MS over the Access Grant Channel (AGCH). It is over the SDCCH that the MS will communicate with the BSS and MSC.
- The MS transmits a location update request message to the BSS over the SDCCH. Included in this message are the MS Temporary Mobile Subscriber Identity (TMSI) and the old Location Area Subscriber (old LAI). The MS can identify itself either with

its IMSI or TMSI. In this example, we will assume that the mobile provided a TMSI. The BSS forwards the location update request message to the MSC.

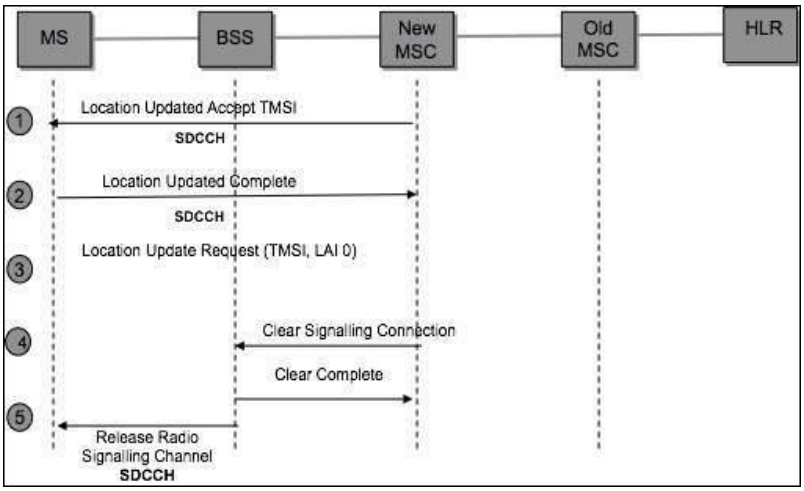
- The VLR analyses the LAI supplied in the message and determines that the TMSI received is associated with a different VLR (old VLR). In order to proceed with the registration the IMSI of the MS must be determined. The new VLR derives the identity of the old VLR by using the received LAI, supplied in the location update request message. It also requests the old VLR to supply the IMSI for a particular TMSI.
- Location Update Scenario-Update HLR/VLR is a point where we are ready to inform the HLR that the MS is under control of a new VLR and that the MS can be de-registered from the old VLR. The steps in update HLR/VLR phase are −
    - The new VLR sends a message to the HLR informing it that the given IMSI has changed locations and can be reached by routing all incoming calls to the VLR address included in the message.
    - The HLR requests the old VLR to remove the subscriber record associated with the given IMSI. The request is acknowledged.
    - The HLR updates the new VLR with the subscriber data (mobiles subscribers' customer profile).



Steps in TMSI Reallocation Phase

- The MSC forwards the location update accept message to the MS. This message includes the new TMSI.

- The MS retrieves the new TMSI value from the message and updates its SIM with this new value. The mobile then sends an update complete message back to the MSC.
- The MSC requests from the BSS, that the signaling connection be released between the MSC and the MS.
- The MSC releases its portion of the signaling connection when it receives the clear complete message from the BSS.
- The BSS sends a "radio resource" channel release message to the MS and then free up the Stand-alone Dedicated Control Channel (SDCCH) that was allocated previously. The BSS then informs the MSC that the signaling connection has been cleared.



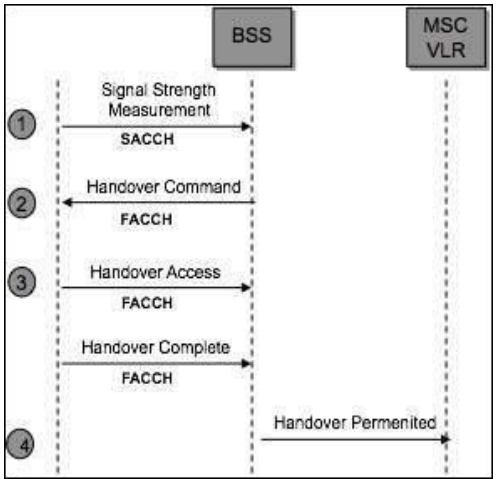
Location Update Periodicity

Location Update automatically takes place when the MS changes its LA. A lot of location updates may be generated if a user crosses LA boundary frequently. If the MS remains in the same LA, Location Update may take place based on time/movement/distance, as defined by the network provider.

Hand Over

This is the process of automatically switching a call in progress from one traffic channel to another to neutralize the adverse effects of the user movements. Hand over process will be started only if the power control is not helpful anymore.

The Hand Over process is MAHO (Mobile Assisted Hand Over). It starts with the Down Link Measurements by the MS (Strength of the signal from BTS, Quality of the signal from BTS). MS can measure the signal strength of the 6 best neighboring BTS downlink (candidate list).



Hand Over Types

There are two types of Hand Over −

- Internal or Intra BSS Handover

Intra-cell hand over

Inter cell hand over

- External or Inter BSS Hand over

Intra-MSC hand over

Inter MSC hand over

Internal handover is managed by the BSC and external handover by MSC.

The objectives of Hand Over are as follows −

- Maintain a good quality of speech.
- Minimize number of calls dropped.
- Maximize the amount of time the mobile station is in the best cell.
- Minimize the number of hand overs.

When will a Hand Over take place?

- Distance (propagation delay) between the MS and BTS becomes too big.
- If the received signal level is very low.
- If the received signal quality very low.
- Path loss situation for the mobile station to another cell is better.

3.9. Personal Area Networks: Bluetooth, Zigbee

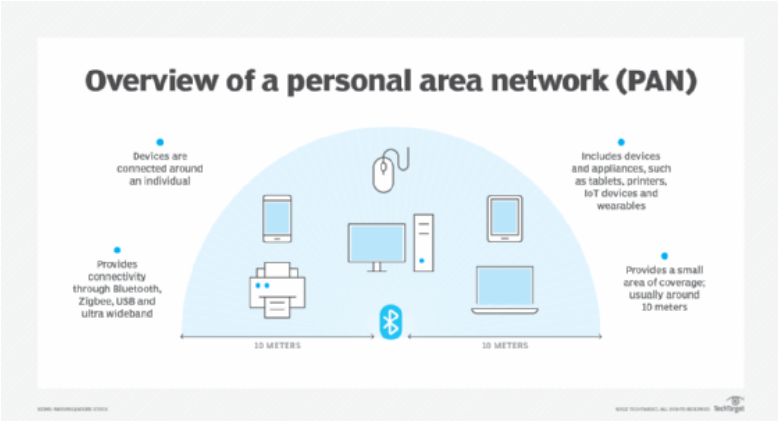
3.9.1. Personal Area Network (PAN)

What is a personal area network (PAN) and how does it work?

A personal area network (PAN) interconnects technology devices, typically within the range of a single user, which is approximately 10 meters or 33 feet. This type of network is designed to enable devices in a small office or home office (SOHO) environment to communicate and share resources, data and applications either wired or wirelessly.

PANs typically consist of laptops, smartphones, tablets, wearables, printers and entertainment devices. These devices are generally interconnected using some form of wireless technology. This kind of PAN could also be connected to the internet or other networks without wires.

The concept of a PAN was first developed by Thomas Zimmerman and other researchers at MIT's Media Lab and was later supported by IBM's Almaden research lab.



Types of PANs

There are two types of PANs: wireless and wired.

Wireless PANs. A wireless PAN is designed to serve a single person, SOHO or small workgroup. As such, limited distance, throughput, peripheral sharing and low volume are some of the main characteristics of this type of network. A wireless PAN functions wirelessly and is carried over a low-powered, short-distance wireless network technology, such as Infrared Data Association (IrDA), wireless universal serial bus (USB), Bluetooth, ultra wideband or Zigbee.

Conceptually, the difference between a PAN and a wireless local area network (LAN) is that the former tends to be centered around one person while the latter is connected without wires and serves multiple users.

Wired PANs. These networks provide short connections between peripherals using wired technologies, such as USB, IEEE-1394 high-performance serial buses or a Thunderbolt hardware interface.

Common use cases for a personal area network

A PAN has several use cases and applications. The following are the most common examples:

Body area network. Also referred to as a wireless body area network or a body sensor network, it's a type of mobile computer network that moves with a person and is typically established when a person connects their smartphone and Bluetooth headphone. With this network, communication takes place entirely within, on and in the immediate proximity of a person. An example of a body area network is a medical sensor that has wireless connections implanted in the human body to monitor vital signs.

Wearable technology. Wearable computer devices, including smart clothing and smart watches with accelerometers, can monitor daily activities and exchange digital information using the electrical conductivity of the human body as a data network.

Offline network. This is also referred to as a home network, as users can establish this type of network in their home offices. Multiple devices, including printers, TVs, game systems, laptops and other home appliances are integrated together through Bluetooth or Wi-Fi, forming a small, single-space network. However, the devices in this network aren't connected to the internet.

SOHO. This type of network is set specifically for work purposes and is separate from other home networks. Using an internet connection, this virtual network is typically a small setup of connected devices that are used solely for office work and may incorporate a virtual private network.

What's the difference between a PAN and a LAN?

PANs and LANs are unique in their own ways. The major difference between these networks is that a PAN connects the devices within the short range of an individual person, whereas a

LAN connects devices at a single site, typically an office building. Similar to a PAN, a LAN can be both wired and wireless.

The following are a few main characteristics of a PAN and a LAN:

Characteristics of a PAN

- A PAN generally provides interconnection of technology devices surrounding a single user.
- Mainly used for low data-rate applications in home automation, it's widely used by
- PAN can include mobile devices and appliances, such as tablets, printers, keyboards, barcode scanners, game consoles, laptops and other personal devices.
- If connected wirelessly to the internet, it's called a wireless personal area network.

Characteristics of a LAN

- The scope of a LAN is usually within a single building or between buildings.
- A LAN is designed for high data transmission rates from megabytes per second to gigabytes per second.
- A Wi-Fi network is an example of a LAN. A LAN is typically established in houses, hospitals and educational institutions. For example, a Wi-Fi network for a university connects all the devices that are connected to it.
- The smallest LAN may only be made up of two computers, while larger LANs can accommodate thousands of computers.
- A LAN typically relies on wired connections for higher speeds and security, but wireless connections can also be part of a LAN.
- LANs are mainly used for their high speed and relatively low cost.

Advantages of PANs

A PAN can be very convenient and easy to set up. The following are a few benefits of using a PAN:

No wires are required. The connecting devices in a PAN only require Bluetooth to be enabled, which eliminates the need for extra wires. This also eradicates the need for cable management and wasted floor space, making it a highly cost-effective network.

Reliable and secure. A PAN network ensures a reliable and stable connection if it's established within the 10-meter range.

Easy data synchronization. A PAN provides easy data synchronization between different devices. As an example, all devices connected within a PAN can be used to exchange, download and upload data with each other.

Portability. A PAN provides extreme portability, as it's wireless, and users can transport devices and exchange data wherever they want.

Disadvantages of PANs

Along with its various advantages, a PAN also has several shortfalls, such as the following:

Short network range and slow data transfer. A PAN uses Bluetooth communication that doesn't span beyond the 10-meter range. This makes long-distance data sharing difficult and slows down the rate of data transfers.

Signal interference. The Bluetooth and IrDA rays used for transmission in a PAN can cause interference with radio signals, which can severely interrupt and degrade the quality of communication between devices.

Cost. Using a PAN can be expensive, as most built-in WPAN devices are costly. Also, most devices used for creating a PAN have a higher price tag, such as smart phones and laptops.

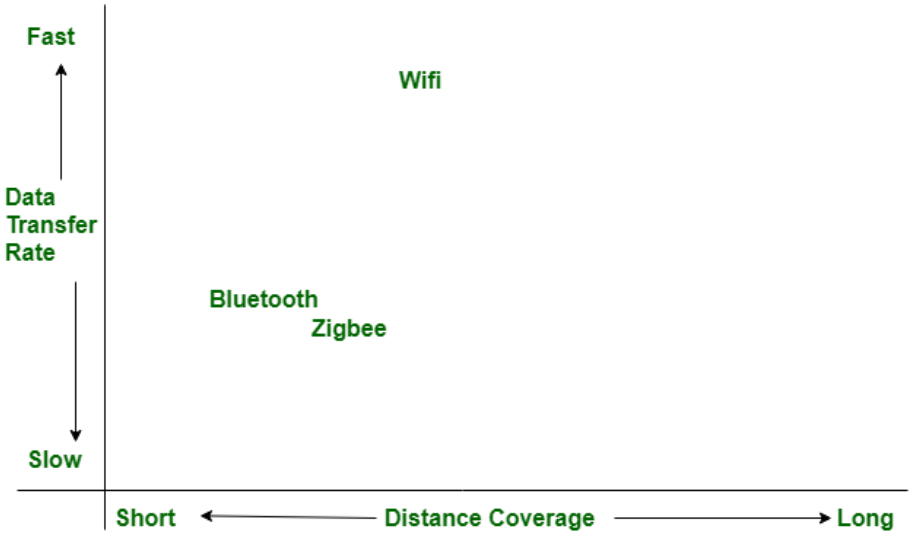
Line of sight propagation. PANs mostly operate on IrDA technology, which travels in a straight line from one point to another, also known as the line of sight propagation. Unlike radio-based communications, IrDA devices must be aligned to work. For example, a TV remote won't work unless it's beamed directly to the TV screen.

Networks can differ based on size, connectivity, design and coverage. Explore the seven common network types, along with their benefits and use cases.

3.9.2. Bluetooth, Zigbee

Bluetooth was developed under IEEE 802.15.1, which is used for providing wireless communication through radio signals. The frequency range supported in Bluetooth varies from 2.4 GHz to 2.483 GHz. It covers less distance than Zigbee. In Bluetooth, GFSK modulation technique is used.

Whereas in **Zigbee**, BPSK and QPSK modulation techniques are used like UWB (Ultra-Wide Band). the frequency range supported in Zigbee is mostly 2.4 GHz worldwide, which means 2.4 GHz is not supported at all times. It covers more distance as compared with Bluetooth.



In the above figure, we can see that the data transfer rate is faster in Bluetooth than in Zigbee whereas Zigbee covers larger distance than Bluetooth.

Both Bluetooth and Zigbee have a lot in common which is, each area unit styles of IEEE 802.15 WPANs. each run within the pair of.4-GHz unlicensed band, and each use tiny kind factors and low power. Besides these similarities, there are some differences which are given below in the tabular form.

S. No.	Bluetooth	Zigbee
1.	The Bluetooth SIG (Special Interest Group) is the organization responsible for managing Bluetooth standards and devices.	The Zigbee Alliance is responsible for managing Zigbee and testing and approving Zigbee-based devices. IEEE standardizes all Zigbee-based protocols.

S. No.	Bluetooth	Zigbee
2.	The frequency range supported in Bluetooth vary from 2.4 GHz to 2.483 GHz.	While the frequency range supported in Zigbee mostly 2.4 GHz worldwide.
3.	There are seventy nine RF channels in Bluetooth.	There are sixteen RF channels in zigbee.
4.	It uses GFSK modulation technique.	Whereas it also uses BPSK and QPSK modulation techniques like UWB.
5.	There is maximum of 8 cell nodes in Bluetooth.	While there is more than sixty five thousand (65000) cell nodes in zigbee.
6.	Bluetooth networks can be built using the point-to-point master-slave approach in which there is one master and up to seven slaves form a piconet, which leads to forming a scatter net which is a linking of two or more piconets.	Zigbee devices can be networked in a variety of generic topologies, including a star, mesh, and others. A cluster can be created by connecting different Zigbee-based network topologies. Zigbee Coordinator, Zigbee Router, and Zigbee Endpoint nodes make up any Zigbee network.
7.	Bluetooth requires low bandwidth.	While zigbee also requires low bandwidth but greater than Bluetooth's bandwidth most of time.
8.	The radio signal range of Bluetooth is ten meters.	While the radio signal range of zigbee is ten to hundred meters.

S. No.	Bluetooth	Zigbee
9.	Bluetooth was developed under IEEE 802.15.1.	Whereas it was developed under IEEE 802.15.4.
10.	Bluetooth batteries may be recharged.	Although ZigBee batteries cannot be recharged, they last longer.
11.	Blue tooth uses high data rates and a lot of power on large packet devices.	Zigbee employs low data rates and little power on small packet devices.
12.	Bluetooth employs the Frequency Hopping Spread Spectrum. In frequency hopping, the carrier signal is made to fluctuate in frequency.	Zigbee employs the Direct Spread Spectrum technique. In direct spread spectrum; the original signal is mixed and recovered from a pseudo random code at transmitter and receiver.
13.	A network speed of up to 250 megabits per second.	A network speed of up to 1 megabit per second.
14.	The time it takes to join a network using Bluetooth is about 3 seconds.	The time it takes to join a network using Zigbee is about 30 milliseconds.
15.	Bluetooth's protocol stack is 250K bytes in size.	Zigbee's protocol stack is 28K bytes in size.
16.	Computer peripherals like wireless keyboards, mice, headsets, and other peripherals are the main use cases for Bluetooth-based applications. Additionally, several wireless remote	Systems built on the Zigbee protocol are intended for wireless sensor networking, and they are more popular with compact and energy-efficient gadgets. Zigbee-based networking is used in a variety of

S. No.	Bluetooth	Zigbee
	controls and gesture-controlled devices communicate data via Bluetooth.	applications, including SCADA system sensors, medical devices, and television remote controls.